

Deval Sonani

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Summary

Experienced Full-Stack Developer with almost **5 years** of experience designing and delivering high-concurrency distributed systems, scalable microservices, and data-intensive backend platforms. Skilled in Java, Go, and Python, with a strong record of leading cross-functional engineering efforts in fast-paced e-commerce and logistics environments. Delivered mission-critical systems from the ground up, including a subscription auto-renewal engine processing **10M+ daily transactions** at GoDaddy and a last-mile load-allocation solver routing **100k+ daily packages** at Delhivery. Known for translating complex technical architectures into clear business outcomes for non-technical stakeholders, and for building teams that ship high-quality software reliably.

Technical Skills

- **Languages:** Java, Python, Go, TypeScript, SQL
- **Backend:** Spring Boot, Django, Flask, FastAPI, Vert.x
- **APIs:** RESTful APIs, GraphQL, Microservices, Hibernate/JPA
- **Frontend:** React, Next.js, Tailwind CSS, Redux, Zustand
- **Databases:** PostgreSQL, PostGIS, DynamoDB, MongoDB
- **Caching & Search:** Redis, Elasticsearch, Couchbase
- **Streaming:** Apache Kafka, AWS Kinesis, AWS Firehose, RabbitMQ
- **Cloud:** AWS S3, AWS EC2, AWS ECS, AWS EKS, AWS RDS, AWS Lambda, AWS Athena
- **DevOps:** Docker, Kubernetes, Helm, GitHub Actions, Databricks
- **Optimization:** Google OR-Tools, Gurobi, Linear Programming, VRP, Uber H3
- **AI/ML:** XGBoost, Scikit-Learn, PySpark, MLflow, LangChain, RAG Pipelines, ChromaDB, LLM APIs
- **Testing:** ArchUnit, Testcontainers, JUnit, Pytest

Experience

Software Development Engineer II (SDE2)

GoDaddy

Mar 2025 – Present

Remote – Pune, India

- Part of a cross-functional team of 15 engineers on the Subscriptions Platform, reporting to the Engineering Manager; collaborated with product, billing, and mobile commerce teams to deliver renewal and dunning systems at scale.
- Designed and authored the HLD/LLD for a multi-region **Auto-Renewal System (ARS)** processing **10M+ daily transactions** using **Java 21** and AWS SQS; modeled state transitions on **DynamoDB Global Tables** to prevent concurrent multi-region race conditions.
- Led the migration of legacy Windows COM+ C++ billing components into a modern **Spring Boot 3.x** microservice, automating failed card retry flows and removing a long-standing source of technical debt.
- Built an asynchronous event sync worker in **Vert.x** to stream MySQL subscription data into a **Couchbase** read-replica cluster on **AWS ECS**, cutting lookup latencies by **40%** and improving dashboard responsiveness.
- Developed an NLP classification pipeline using **Gemini LLM** and **ChromaDB** vector embeddings to categorize payment declination logs; configured A/B experiments in the dunning-policies-service to optimize customer recovery rates.
- Refactored Apple and Google in-app purchase (IAP) validation pipelines with cryptographic signature verification and idempotent transaction logging; authored API documentation and operational runbooks adopted across the team.
- Elevated engineering standards by introducing **ArchUnit** architectural tests and dockerized **Testcontainers**; mentored three junior engineers through structured code and design reviews, raising test coverage by **15%**.

Software Development Engineer II (SDE2)

Cars24

Mar 2024 – Mar 2025

Gurugram, India

- Served as senior engineer for a team of 8 engineers, reporting to the VP of Engineering; drove the architecture and delivery of a new enterprise **HRMS platform** built on Java/Spring Boot deployed on **AWS ECS** and **AWS RDS**.
- Decomposed a tightly coupled legacy monolith into independently deployable microservices, optimizing cloud resource provisioning and reducing AWS infrastructure costs by **25%**.
- Designed the organizational hierarchy data model using the Closure Table pattern in **PostgreSQL** with multi-level **Redis** caching, accelerating org-tree and budget-rollup queries from **4.2 seconds to 35 milliseconds**.

- Communicated system architecture and tradeoffs directly to C-suite stakeholders, translating technical decisions into business value; automated payroll data workflows that saved the finance team **20%** in monthly reconciliation time.
- Delivered a high-concurrency real-time notification gateway in **Go (Golang)** using goroutines and WebSockets, enabling instant payroll and HR alerts to **10,000+ employees** with sub-second delivery.

Software Development Engineer I (SDE1)

Jan 2022 – Mar 2024

Delhivery

Gurugram, India

- Designed and built **Flare**, Delhivery's core last-mile load-allocation engine in Django/Python, modeling driver-to-shipment assignments as a **Mixed-Integer Linear Program** via SCIP/Gurobi, handling **100k+ packages daily** across **500+ distribution centers** with capacity and shift constraints.
- Built a **Vehicle Routing Problem (VRP)** layer using affinity matrices to compute optimal delivery zones per driver; implemented **Uber H3 spatial indexing** to auto-generate dispatch territories, improving last-mile utilisation by **25%**.
- Developed a high-concurrency **FastAPI/asyncio** microservice with a custom C/C++ compiled module for spatial polygon computation; engineered the **AVTD preprocessing pipeline** using Elasticsearch DSL to filter out geocoding and volumetric errors before solver ingestion.
- Tuned **PostGIS** spatial indexes and rewrote complex polygon-join queries, cutting geographic lookup latency by **50%**; automated 50GB+ daily CSV processing with Celery and stored results in **AWS S3** and **AWS Athena**.
- Architected a **Java/Spring Boot** event-streaming gateway publishing real-time allocation updates via **Apache Kafka** and **AWS Kinesis**; containerized and deployed services on **AWS EKS**.
- Delivered ETL pipelines in **Databricks (PySpark)** processing TB-scale GPS logs; trained and deployed **XGBoost** models via MLflow for fleet capacity prediction with **88% accuracy**.
- Authored system design wikis, onboarding documentation, and internal runbooks for the solver platform, significantly reducing ramp-up time for new engineers; placed **Runner-Up** at Google x Delhivery Gen-AI Hackathon for an RAG-powered logistics support agent.

Projects

Nanoscale MESFET Semiconductor Modeler

ATLAS TCAD, MATLAB, Numerical Analysis

Research Project – DA-IICT

- Simulated nanoscale depletion-mode MESFETs using ATLAS TCAD, modeling carrier transport equations, gate length scaling, and junction field profiles to study device behavior at the nanoscale.
- Analyzed the effects of parameter variations including channel doping depth, barrier height, and gate width on I-V transfer characteristics to optimize saturation currents and pinch-off thresholds.
- Co-authored and published "*Analysing the Effects of Varying Parameters in a MESFET*" in IEEE – IEEE Link.

Education

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT)

Jul 2018 – May 2022

Bachelor of Technology in Computer Science | CGPA: 8.8 / 10

Gandhinagar, Gujarat

Achievements, Certifications & Interests

Certifications: AWS Solutions Architect Associate; MongoDB Developer Associate; attended AWS re:Invent.

Hackathon: Runner-Up – Gen-AI Hackathon (Google x Delhivery) – Built an AI-powered logistics query agent using RAG and LLMs, placing 2nd out of 40+ teams.

Academic: ACPC All-India Rank 24 – Admission Committee for Professional Courses, Gujarat.

Publication: Co-authored "*Analysing the Effects of Varying Parameters in a MESFET*" – IEEE, 2022.

Competitive Programming: Active on Codeforces and LeetCode; focused on graph algorithms, dynamic programming, and system-level optimization problems.

Interests: Open-source contributions, distributed systems architecture, and IoT home automation projects.